# Manish Agarwal

LinkedIn GitHub LeetCode

## Education

**National Institute of Technology Karnataka, Surathkal** — Surathkal, Karnataka
*BTech in Computer Science and Engineering* | ***CGPA 8.02*** — *Aug 2023 – Present*

## Experience

**Software Development Engineer Intern** — Hyderabad, India
*Amazon* — *May 2025 – Jul 2025*

- Built a **black-box regression harness** for a rewrite of a money-critical refund library, verifying output parity with production to **4 decimal places** across **76 test cases** in 3 layers, **before deployment**.
- Solved the blocker of logs holding only inputs and outputs: built a **propose-verify-repair generator** that reconstructs missing upstream values and keeps only test cases that **reproduce production output exactly**.
- Integrated an **AI agent** that re-wires tests after refactors but cannot edit expected outputs; validated with an injected currency bug that the harness caught while unit tests stayed green.

## Projects

**SwiftShareX** | | | *React Native, C++/JSI, TCP Sockets, UDP Discovery* — Jun 2025 – Dec 2026

- Built an offline file-sharing app across **3 platforms** (Android, macOS, Windows) using **React Native** and a native **C++ TCP engine**, sharing **one UI codebase** with Electron and reaching **80 MB/s** over local Wi-Fi.
- Architected a custom binary protocol (SWFT) streaming **256 KB chunks** through **8 MB socket buffers**, with receiver-confirmed delivery and atomic writes, so **multi-GB files** are never falsely marked sent or left corrupted.
- Integrated **JSI** to expose a multi-threaded C++ engine directly to JavaScript, polling live progress every **250 ms** without bridge overhead, and streamed files from Android file descriptors via **JNI** without intermediate copies.
- Implemented **UDP broadcast discovery** with a **6-message** control protocol for pairing, multi-file queues and per-file cancel, securing receivers with a peer IP allowlist and **4 MB** chunk cap.

**SmartGalleryX** | | *PyTorch, SigLIP 2, FastAPI, React Native, SQLite* — Jun 2026 – Nov 2026

- Fine-tuned a **SigLIP 2 ViT-L/16** model on **146K images and 730K captions** with sigmoid contrastive loss and multi-GPU DDP, enabling natural-language search over phone photos.
- Raised top-1 retrieval from **84.3% to 90.1%** on Flickr30k and **55.0% to 64.0%** on COCO, fitting ViT-L on 16 GB GPUs with layer-wise LR decay and gradient checkpointing.
- Built a **stateless FastAPI** embedding server with per-GPU encoders and **int8 quantization** (**4x smaller** vectors), API-key auth and caching, embedding **30+ images/s**.
- Developed a React Native app with an **on-device vector index** and **min-heap top-k ranking** that syncs only new or edited photos and serves repeat searches **offline**.

## Technical Skills

**Languages:** C/C++, Swift, JavaScript, Python, MySQL, HTML/CSS
**Frameworks/Libraries:** SwiftUI, React Native, React, Firebase, Tailwind CSS, NumPy, Pandas
**Tools:** Xcode, Android Studio, VS Code, Git, Postman, Expo CLI, Kaggle

## Achievements

- Secured **AIR 5085** in *JEE Advanced* and **AIR 2645** in *JEE Main.*
- Secured **1st place** out of **200+ teams** in *Silicon Maze Capture the Flag*, organized by WebClub, NITK.
- Achieved **2nd place** out of **120+ teams** in *Code Red Capture the Flag*, hosted by IEEE, NITK.

## Extracurricular

**Frontend Lead** – **ACM NITK Website** | *React.js, UI/UX Design* — Feb 2025 – Present

- Leading the frontend team for the official ACM NITK website, focusing on **UI/UX consistency**, **responsive design**, and **accessibility**.
- Implemented and maintained critical React.js components, improving user interaction and performance.